Forecasting Demand, Not Sales: A Practitioner's Guide to Probabilistic Models for Stockout-Censored Data

Juan Orduz
juanitorduz@gmail.com

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Abstract

Recorded sales are a censored measurement of demand: stockout days record zeros and capacity days record caps, and forecasting models fit to raw sales learn the censoring as if it were demand, feeding under-orders that cause the next stockout. This practitioner's guide presents three probabilistic mechanisms for forecasting demand rather than sales, selected by what the data records about each stockout. A binary on-shelf flag admits *gating*: an intermittent-demand smoother in the Croston and Teunter-Syntetos-Babai family freezes its occurrence updates during off-shelf periods. A per-observation censoring point admits a *censored likelihood*: capped observations enter a Tobit-style model as lower bounds, which halves peak-day errors in simulation exactly where a plain likelihood saturates at the cap. A fractional, noisily labeled availability admits a *multiplicative saturating factor with a learned floor*, demonstrated on the FreshRetailNet-50K dataset, where the full 50,000-series hierarchical model fits in under six minutes of stochastic variational inference on a GPU. All three share one structure: the corruption is a switchable model component, and the demand forecast is the model with the component switched off. We pair the mechanisms with an evaluation discipline for censored settings, showing concretely that aggregate accuracy against recorded sales rewards reproducing the corruption, and we state the limitations of each mechanism explicitly, including exogenous-availability assumptions, Gaussian misspecification, and the absence of cross-mechanism comparisons. All case studies are reproducible from the open-source numpyro_forecast package.

1 Introduction

Every demand forecasting system trained on retail data faces the same quiet corruption: the training data records *sales*, but the decisions it feeds, ordering, assortment, replenishment, need *demand*. The two diverge exactly when it hurts. On a stockout day the till records zero however many customers wanted the item; on a day the shelf ran out at noon it records half a day's appetite; on a day demand exceeded the delivered quantity it records the delivery. A model fit to raw sales learns these artifacts as if they were demand patterns: it learns that the item "does not sell" after stockout runs and that peak demand "tops out" at shelf capacity. The forecast then feeds the ordering decision, the low forecast produces a low order, and the low order produces the next stockout. Yesterday's stockouts are baked into tomorrow's orders.

The repair is conceptually uniform even though its implementations differ. Treat the latent demand as the modeled quantity, treat whatever is recorded about availability as data, and make the corruption an explicit, switchable component of the joint model. The sales forecast is the model with the corruption component on; the demand forecast, the object planning actually needs, is the

Symbol	Meaning	Domain
d_t	latent demand	$[0, \infty)$
y_t	observed sales	$[0, \infty)$
a_t	availability	$\{0, 1\}$ (§3, §4) or $[0, 1]$ (§5)
c_t	censoring point (capacity, inventory level)	$(0, \infty)$, observed per period in §4
ℓ_t	a smoothed or state-space level	model specific
g_t	update gate of a smoothing recursion	$\{0, 1\}$
$f(a_t)$	multiplicative availability factor	$(0, 1]$
h	forecast horizon	positive integers

Table 1: Notation shared by the three case studies.

same model with the component switched off. What varies across situations is which corruption component the recorded data can support, and that is the organizing question of this paper: *what do you observe about the stockout?*

This paper is a practitioner's guide built around three worked case studies, one per answer to that question, all implemented in the open-source numpyro_forecast package and reproducible from its example notebooks (§7). When only a *binary on-shelf flag* is recorded, an intermittent-demand smoother can gate its occurrence updates on availability, so off-shelf zeros stop eroding the demand estimate (§3). When a *per-observation censoring point* is recorded, a Tobit-style censored likelihood keeps capped days in the fit as lower bounds instead of corrupted values (§4). When availability is *fractional and noisily labeled*, a multiplicative saturating factor with a learned floor attaches to a hierarchical state-space model and survives label noise at a scale of 50,000 series (§5). Alongside the mechanisms, the case studies apply a common evaluation discipline (§2.3): aggregate accuracy against recorded sales rewards models that reproduce the corruption, so demand models must be judged on calibration, stress-day behavior, and counterfactual plausibility instead, and one case study exhibits the trap concretely by showing the mechanism-free model winning every point metric against sales while halving its accuracy advantage into a deficit on the days demand exceeded supply.

The scope is deliberately bounded. This is not a benchmark: no mechanism is applied to another's data situation, so the paper characterizes mechanisms and their failure modes rather than ranking them, and the controlled comparison that a ranking would require is spelled out in §6. None of the mechanisms is claimed as new; gating, censored likelihoods, and attenuation factors all have long histories reviewed in §2.4, and the availability-gated smoother of §3 in particular is a one-gate modification of the method of Teunter et al. (2011). The contribution is the unified presentation, the honest accounting of what each mechanism does and does not buy on worked examples with stored, verifiable outputs, and the demonstration that the whole toolkit runs at production scale in a modern probabilistic-programming stack.

2 Sales as censored demand

2.1 Notation and the demand forecast

Throughout the paper, $d_t \geq 0$ denotes the latent demand for one item at time t: the quantity customers would have bought had the item been fully available. The recorded sales y_t are a corrupted measurement of d_t. The corruption is driven by an availability signal a_t and, where one exists, a censoring point c_t (a shelf capacity or an inventory level). In a panel, a subscript s indexes the series (a store and product pair). Table 1 collects the symbols; all three case studies use them with the same meaning.

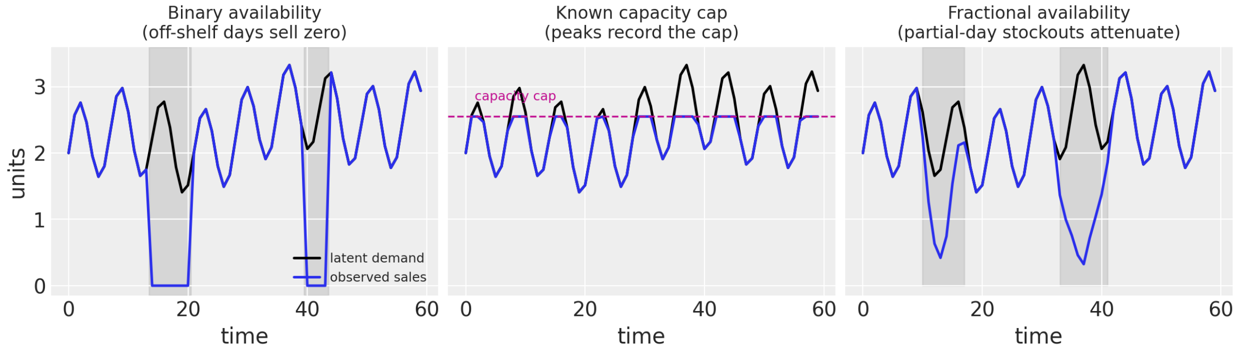

Figure 1: Three ways stockouts corrupt the same latent demand path (black). Left: a binary availability flag gates sales (blue) to zero during off-shelf runs (shaded). Center: a known capacity cap records $\min(d_t, c)$, so peak observations are lower bounds. Right: fractional within-day availability attenuates sales toward a small floor. Each column corresponds to one case study in Table 2.

The object every method in this paper targets is the *demand forecast*: the predictive distribution of what will be demanded over the horizon, not of what will be rung up at the till. All three case studies obtain it by the same meta-move: the corruption enters the joint model as an explicit, switchable component, and the demand forecast is the model's predictive distribution with that component switched off. How the switch is thrown differs by mechanism. In the two availability-driven case studies it is a covariate edit: a full-availability scenario input in §3 and availability pinned to one over the horizon in §5. In the capacity case study of §4 it is a structural omission: the forecast recursion simply never applies the cap, so the censoring mechanism shapes the demand forecast only indirectly, through the parameter posterior and a lag filter. The distinction matters when reading Table 2, whose final column is deliberately not uniform.

2.2 A taxonomy of stockout information

The organizing question of this paper is not which model is best. It is: *what do you actually observe about the stockout?* The answer determines which corruption mechanisms can enter the likelihood at all. A binary on-shelf flag admits gating a smoothing recursion. A censoring point recorded per observation admits a censored likelihood. A fractional, noisily labeled availability admits a multiplicative attenuation factor whose floor absorbs the label noise. Figure 1 shows the three corruption patterns on a common toy demand path, and Table 2 maps each pattern to its mechanism and to the case study that works it out. The table is adapted from the taxonomy developed in the numpyro_forecast example collection.

Two scope notes keep this table honest. First, the rows are a statement about *availability of mechanisms*, not about optimality: no mechanism is cross-applied to another row's data situation in this paper, and two of the six cross-applications are structurally infeasible because a censored likelihood requires a recorded censoring point, which neither the binary panel of §3 nor the fresh-retail data of §5 contains. Second, where a within-case ablation exists (§3 and §4), the mechanism is compared against the *same* model with the mechanism switched off, and the improvement shows up on specific criteria stated in each case study, not uniformly across all metrics.

Stockout information observed	Censoring pattern	Mechanism	Demand forecast	§
Binary on-shelf flag $a_t \in \{0, 1\}$	hard zeros while off shelf	gate the recursion updates on a_t; mask the likelihood	full-availability scenario input over the horizon	§3
Per-observation censoring point c_t	observations record $\min(\cdot, c_t)$	censored likelihood: density below the cap, survival mass at it	generative recursion that never applies the cap	§4
Fractional, noisy availability $a_t \in [0, 1]$	multiplicative attenuation, mislabeled hours	saturating factor $f(a_t)$ with a learned floor	availability pinned to one over the horizon	§5

Table 2: Selection map from observed stockout information to censoring mechanism. The mechanisms compose; the rows mark which signal makes each mechanism *available*, not which is optimal for a given dataset (see the scope note below).

2.3 What to score a demand forecast against

Evaluating a demand model on stockout-affected data has a trap built into it. On censored days the recorded sales are precisely the quantity the model is designed *not* to reproduce. A model that correctly infers that demand exceeded the cap will, on those days, sit above the observations and be penalized for it; a model that happily reproduces the corruption will score well. Aggregate accuracy against recorded sales is therefore partly an evaluation of the censoring process, not of the demand model. This is not a hypothetical concern: §4.4 shows a case where the mechanism-free model wins every point metric against observed sales while systematically under-ordering exactly on the days that drive stockouts.

What should be used instead depends on what is observable. On simulated data the latent demand is known, and scoring against it is the point of the simulation. On real data the latent demand is never observed, and we fall back on three weaker but workable criteria: *calibration* of the predictive intervals, *behavior on stress days* (days where demand plausibly exceeded supply), and *counterfactual plausibility* (whether the implied full-availability forecast is consistent with independent evidence such as the empirical demand-availability curve). Each criterion comes with caveats that the case studies state explicitly. In particular, interval coverage on short holdouts is noisy: on a 30-day test window the standard error of an empirical coverage estimate at nominal 50% is roughly $\sqrt{0.5 \cdot 0.5/30} \approx 0.09$, so coverage differences below about 0.2 should be read as directional at best. Throughout, reported coverage refers to central (equal-tailed) intervals, while the figures display highest-density intervals; for skewed predictive distributions the two genuinely differ.

The synthetic case studies (§3 and §4) should be read as *mechanism validation*: can the model recover a known latent quantity from corrupted observations? In neither case is the fitted model the data-generating process. Case study 1 fits Gaussian smoothing recursions with noise floors to Gamma-Poisson count data and a binary indicator; case study 2 omits the generator's friction term. Both models are misspecified in ways that work against the mechanism rather than for it, so recovery is not guaranteed by construction. The framework these studies test is falsifiable: it would be undermined if the mechanism-free variant recovered the latent demand quantities equally well, or if the mechanism-aware variant lost on stress days and calibration simultaneously.

Finally, a comparability warning that applies to every table in this paper. The three case studies use different data, different scales, different likelihoods, different inference algorithms (mean-field variational inference in §3 and §5, Hamiltonian Monte Carlo in §4), and different evaluation targets (observed panel sales, latent simulated demand, rescaled clipped sales). CRPS and coverage values are therefore not comparable across case studies, and each results table repeats this warning where it applies.

2.4 Related approaches not covered here

The mechanisms in this paper sit inside a large literature on demand estimation under censoring, and several important strands are deliberately out of scope. The revenue-management community calls the problem *unconstraining*: recovering true demand from booking data truncated by capacity controls (Talluri and van Ryzin, 2004; Weatherford and Pölt, 2002; Queenan et al., 2007). The inventory literature studies parametric demand estimation from sales in lost-sales systems, from early work on Poisson demand through censored newsvendor formulations (Conrad, 1976; Nahmias, 1994; Agrawal and Smith, 1996; Ding et al., 2002); Braden and Freimer (1991) established how censored observations change Bayesian updating, a result that matters here because all three of our case studies treat availability as exogenous, an assumption that fails when stockouts correlate with demand. A separate strand estimates *primary* demand under stockout-based substitution with discrete-choice models, which answers a question our single-item models cannot: where did the lost demand go (Anupindi et al., 1998; Kök and Fisher, 2007; Musalem et al., 2010; Vulcano et al., 2012). Data-driven newsvendor methods handle censoring inside the decision problem rather than the forecast (Sachs and Minner, 2014; Huber et al., 2019). Finally, the FreshRetailNet-50K release that supplies the data for §5 ships its own neural baselines for latent demand recovery (Wang et al., 2025); we do not compare against them because our target there, the full-availability demand forecast, is not scoreable against their observed test sales, as §5 discusses. Readers whose stockout problem involves substitution, endogenous availability, or decision-integrated estimation will find better starting points in these literatures.

3 Case study 1: binary availability, gating the recursion

Intermittent demand series, where many periods record zero, are the natural home of the first mechanism, because a stockout manufactures exactly the symptom that intermittent-demand methods are built around. The classical estimators of this family are all exponential smoothing recursions of one common form,

$$\ell_t = \ell_{t-1} + g_t\,\alpha\,(x_t - \ell_{t-1}), \tag{1}$$

where x_t is the smoothed signal, $\alpha \in (0,1)$ is a smoothing weight, and $g_t \in \{0,1\}$ is an update gate. The methods differ only in what they smooth and when the gate opens. This section presents Croston's method, the Teunter-Syntetos-Babai (TSB) variant, and an availability-gated variant of TSB in that single frame, so that the modification handling stockouts is visible as a one-line change of gate. The availability-gated variant is a natural modification of TSB rather than a new estimator; versions of it appear whenever practitioners restrict occurrence updates to on-shelf periods, and the contribution here is the unified presentation and a controlled comparison on a common panel.

3.1 Croston and TSB

Croston's method (Croston, 1972) decomposes an intermittent series into demand sizes and inter-demand intervals. Two recursions of the form (1) run side by side, both gated on demand events ($g_t = \mathbf{1}[y_t > 0]$): one smooths the positive demand sizes z, the other the (inverse) interval between

Single simulated series, $n = 80$, one-step-ahead CV (12 folds); not comparable to Table 4 or later tables.				
Method	CRPS (one-step)	CRPS (fixed origin)	Coverage 50%	Coverage 94%
Croston	0.348	0.373	0.08	1.00
TSB	0.228	0.244	0.58	1.00

Table 3: Croston versus TSB on one simulated intermittent series (no stockouts involved yet). Croston's 0.08 coverage at nominal 50% is structural: its central interval is a rate interval lying strictly above zero on a mostly-zero test span. TSB's higher coverage is partly produced by Gaussian predictive mass below zero.

demand events, and the forecast rate is $\hat{y}_{t+h} = \hat{z}_t \, \widehat{1/p}_t$, the size estimate times the inverse-interval estimate. Because $\mathbb{E}[1/p] > 1/\mathbb{E}[p]$, this construction is biased upward; the correction of Syntetos and Boylan (2005) multiplies the rate by $(1 - \alpha/2)$, and Syntetos and Boylan (2001) analyzes the bias itself. In our Bayesian implementation both smoothing weights carry $\text{Beta}(2, 20)$ priors, the initial levels $\text{Normal}(0, 1)$ priors, and the one-step-ahead likelihoods are $\text{Normal}(\ell_{t-1}, \sigma)$ with $\sigma \sim \text{HalfNormal}(1)$, fit with the No-U-Turn Sampler (Hoffman and Gelman, 2014). On a simulated Poisson(0.3) series of 80 periods (73% zeros, 12-fold one-step-ahead cross-validation), the posterior mean forecast rate is 0.699, corrected to 0.663 by the Syntetos-Boylan factor, against a training-window mean demand of 0.338; these are rates, not accuracy scores. Full derivations and diagnostics are in Boylan and Syntetos (2021) and the accompanying notebooks.

Croston's known blind spot is obsolescence: between demand events nothing updates, so a series that dies keeps its last forecast forever. TSB (Teunter et al., 2011) fixes this by replacing the interval channel with a demand *probability* channel that smooths the occurrence indicator $d_t = \mathbf{1}[y_t > 0]$ on *every* period ($g_t \equiv 1$):

$$\ell^p_t = \ell^p_{t-1} + \alpha_p \, (d_t - \ell^p_{t-1}), \tag{2}$$

with forecast $\hat{y}_{t+h} = \hat{z}_t \, \hat{p}_t$. In the implementation the two methods share the same level-recursion code; TSB is obtained by passing an always-open gate to the probability channel. Table 3 reports both methods on the same single series under the same protocol. TSB scores better there, but two caveats from the source notebooks temper the comparison: on this stationary series TSB's headline advantage (probability decay under obsolescence) never activates, and part of its better empirical coverage comes from the Gaussian probability channel placing predictive mass below zero, which a Bernoulli channel would handle more honestly. Croston's near-zero 50% coverage is structural rather than an inference failure: its predictive interval is an interval for a *rate*, strictly above zero, evaluated on a test span dominated by zero counts.

3.2 The availability gate

Now let stockouts enter. When an item is off shelf, $y_t = 0$ regardless of demand. TSB's probability channel (2) treats every such zero as evidence that demand is fading and decays $\hat{p}_t$ toward zero; a long stockout run drives the demand probability, and with it the forecast, into the ground. This is the mechanism by which yesterday's stockouts become tomorrow's under-orders. If a binary availability flag a_t is recorded, the fix stays within the frame of (1): gate the probability channel on availability, $g_t = a_t$, so that off-shelf periods neither confirm nor deny demand, and mask the corresponding likelihood contributions so off-shelf indicators carry zero log-density. The demand-size channel keeps its Croston gate ($y_t > 0$ implies $a_t = 1$, so sizes are only ever observed on shelf). The observation model is $y_t = a_t \, d^*_t$: availability censors demand multiplicatively. For the forecast, availability becomes a scenario input: expected sales over the horizon are $\hat{y}_{t+h} = a_{t+h} \, \hat{z}_t \, \hat{p}_t$ under an assumed availability path, and the demand forecast is the same expression with the full-availability path $a_{t+h} \equiv 1$. Under this gating, $\hat{p}$ estimates $P(\text{demand} \mid \text{available})$, whereas plain TSB estimates the product $P(\text{available}) \cdot P(\text{demand} \mid \text{available})$: a quantity contaminated by the availability process.

3.3 Setup

The test bed is a simulated panel of 1,000 series over 60 periods (train 50, test 10). Per-series demand rates are drawn $\lambda_i \sim \text{Gamma}(2.5)$, latent demand $d^*_{t,i} \sim \text{Poisson}(\lambda_i)$, availability $a_{t,i} \sim \text{Bernoulli}(0.6)$ independently of demand, and observed sales are $y_{t,i} = a_{t,i}\, d^*_{t,i}$. Half of all periods record zero sales; a 0.40 share of periods are stockouts and 0.11 are genuine no-demand periods on shelf. The model is fit with stochastic variational inference (Hoffman et al., 2013) using an `AutoNormal` mean-field guide (Kucukelbir et al., 2017), Adam with learning rate 0.001, and 10,000 steps over roughly 4,000 latent dimensions; smoothing weights carry $\text{Beta}(1.5, 3)$ priors and the size channel a hierarchical noise scale. Two implementation details are honesty-relevant. First, small constant noise floors (0.1 on sizes, 0.05 on the indicator) are added to stabilize SVI, which otherwise produces numerical failures; they are a pragmatic device, not part of the statistical story. Second, the model is deliberately misspecified relative to the generator: Gaussian likelihoods smooth integer counts and a binary indicator, so parameter recovery is not guaranteed by construction (§2.3).

The comparison model requires no second implementation: because availability enters only through the gate and the likelihood mask, refitting the same model with an all-ones availability input recovers plain TSB exactly. This makes the comparison a clean within-model ablation, and it is best read as a consistency check on the mechanism rather than a horse race between different methods.

3.4 Results

Figure 2 shows the mechanism at work on a single series with a 9-period stockout run: the gated posterior rides through the gap unchanged while the ungated recursion decays toward zero and has to recover afterwards. Figure 3 shows the same effect panel-wide at the parameter level. The availability-gated model's estimated demand probabilities line up on the identity against the true per-series probabilities, with a panel mean of 0.99 of truth; the plain-TSB refit lines up on the 0.6 line, with a panel mean of 0.60 of truth. The value 0.60 is not incidental: it is the generator's availability rate, reproduced arithmetically because plain TSB estimates $\mathbb{E}[a_t d_t] = 0.6\, \mathbb{E}[d_t]$. The figure therefore confirms that the bias of ignoring availability has exactly the predicted sign and magnitude; it is a consistency check with a known answer, not a measured performance gap on an open question.

The practical consequence appears when the model is asked the planning question: what would be sold under full availability? Summed over the panel, the availability-gated demand forecast totals 2,481.3 expected units per period against a true total expected demand of 2,471.8, while the plain-TSB refit totals 1,479.8, which is 60% of the truth (Table 4). Two framing points keep this result in proportion. First, under the *historical* availability regime the two models' sales forecasts are much closer, because plain TSB's bias partly cancels when the future looks like the corrupted past; the bias bites specifically on the counterfactual scenario question that assortment and purchasing decisions ask. Second, on realized-availability test sales the gated model's panel CRPS is 0.547 with central-interval coverage 0.70 at nominal 50% and 0.98 at nominal 94%; the mean-field guide and Gaussian likelihoods leave the intervals imperfectly calibrated, and we report these numbers as context rather than as a claim of superiority.

Figure 4 shows the single most decision-relevant behavior: for a series whose last five training

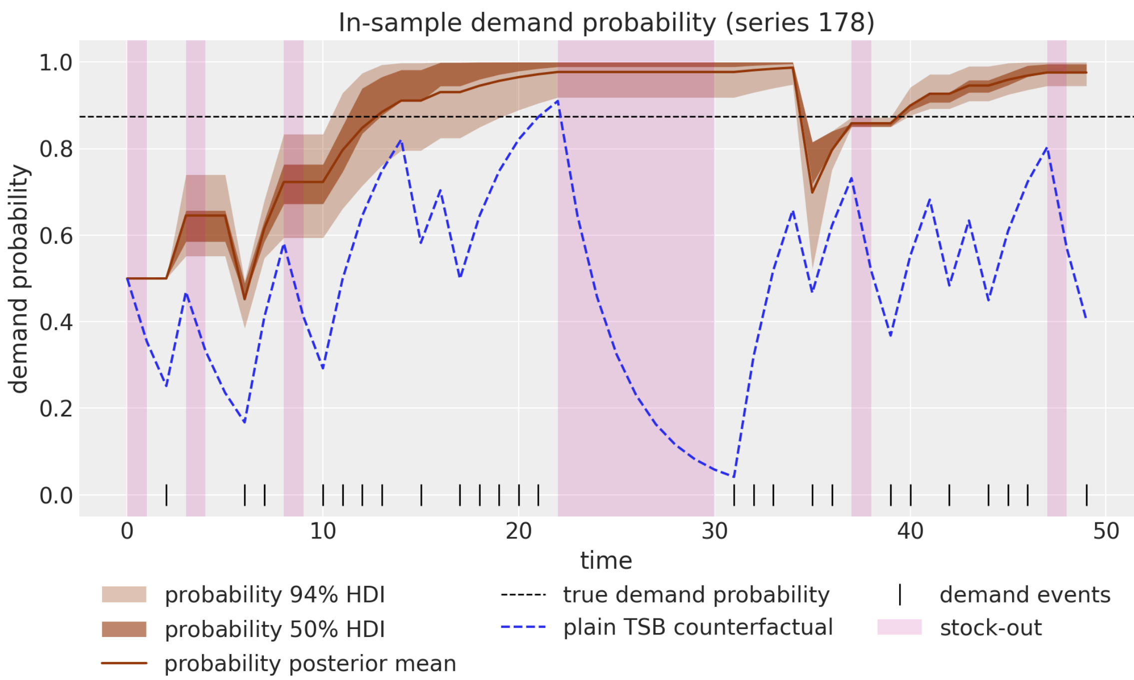

Figure 2: In-sample demand probability for one series with a 9-period stockout run (shaded bands). The availability-gated posterior (solid, with 50% and 94% HDI bands) freezes through stockouts and tracks the true demand probability (dashed horizontal line). The blue dashed path is a deterministic *plain-TSB counterfactual recursion*: the TSB update replayed with the fitted posterior-mean smoothing weight, not the refit plain-TSB model of Table 4. It decays toward zero through every stockout run and must climb back afterwards.

periods are all stockouts (with an estimated 33 units of demand lost during the run), the full-availability forecast opens at the inferred demand level rather than at the recent zeros. A model without the gate starts such series near zero, and the error compounds through the reorder cycle.

Limitations. The Gaussian likelihoods are a pragmatic simplification: predictive draws go negative for counts and outside $[0, 1]$ for the probability channel, and the noise floors exist to stabilize SVI rather than to model anything. Availability is treated as exogenous; when stockouts correlate with demand, as they do for best-sellers, the censoring is informative and the frozen update, while far better than the decaying one, is no longer the full story (Braden and Freimer, 1991). Ground-truth scoring of demand probabilities is possible only because the data are simulated. All panel results use a single fixed train-test split, and the mean-field variational posterior is known to understate uncertainty.

4 Case study 2: known caps, censored likelihoods

The second data situation is richer than a binary flag: for each period we know not only whether the item was constrained but *at what level.* Shelf capacity, delivered quantity, or a purchase limit gives a per-observation censoring point c_t, and days at the cap record a *lower bound* on demand rather than a corrupted zero. That extra information admits a sharper tool than gating: a censored likelihood that keeps capped observations in the fit while scoring them correctly as bounds.

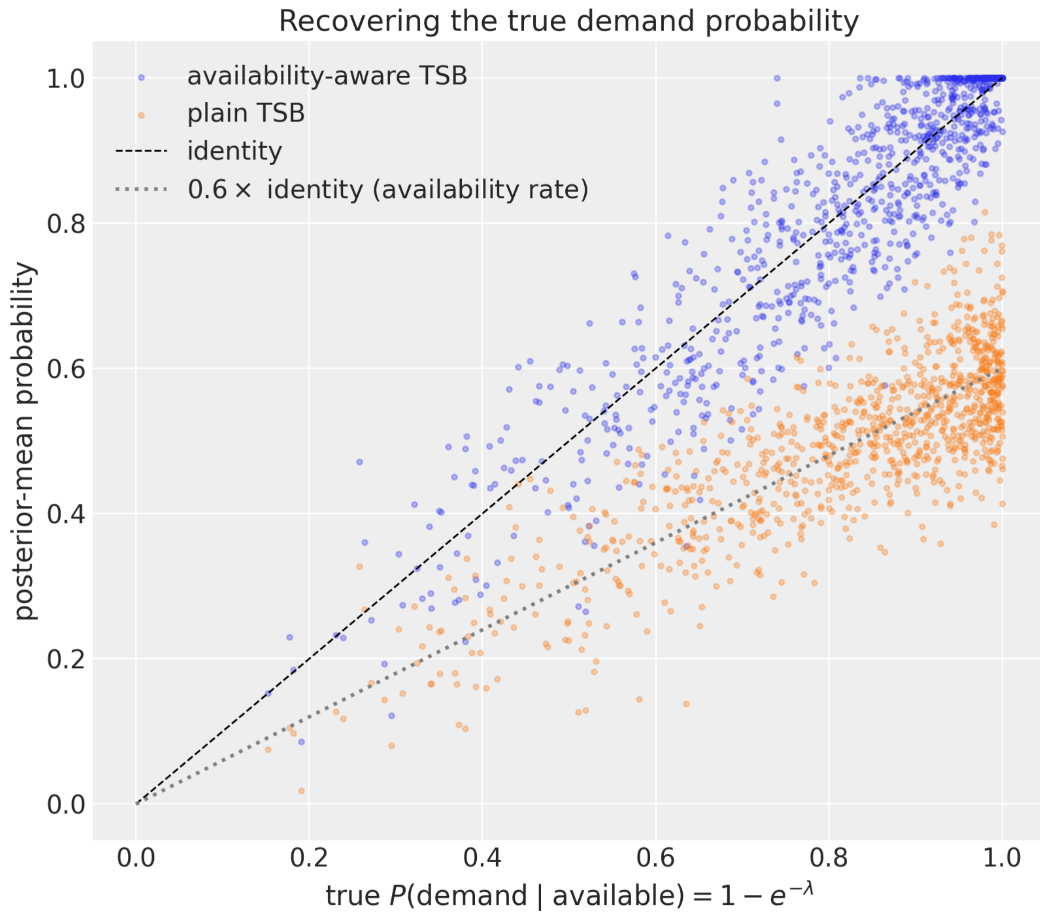

Figure 3: Estimated versus true demand probability across the 1,000-series panel. The availability-gated model (identity line) recovers the conditional demand probability; the plain-TSB refit recovers 0.6 of it, the generator's availability rate, because it estimates the product of availability and demand probabilities.

4.1 The Tobit observation model

Right-censored regression goes back to Tobin (1958); Amemiya (1984) reviews the model family. Let $\hat{y}_t$ be the model's conditional mean and c_t the known cap, with censoring indicator $c_t^{\text{ind}} = \mathbf{1}[y_t \geq c_t]$. The observation density is

$$p(y_t \mid \hat{y}_t, \sigma) \;=\; \text{Normal}(y_t \mid \hat{y}_t, \sigma)^{1-c_t^{\text{ind}}} \left[1 - \Phi\left(\frac{y_t - \hat{y}_t}{\sigma}\right)\right]^{c_t^{\text{ind}}}, \tag{3}$$

so uncensored days contribute the usual density and capped days contribute the survival probability of exceeding the cap. In NumPyro this is a single vectorized observation site using `RightCensoredDistribution`, whose sampler draws from the *uncensored* base distribution, so posterior predictive draws live on the demand scale automatically. Days with the item fully off shelf carry no demand information at all and are removed from the likelihood with a mask, exactly as in §3.

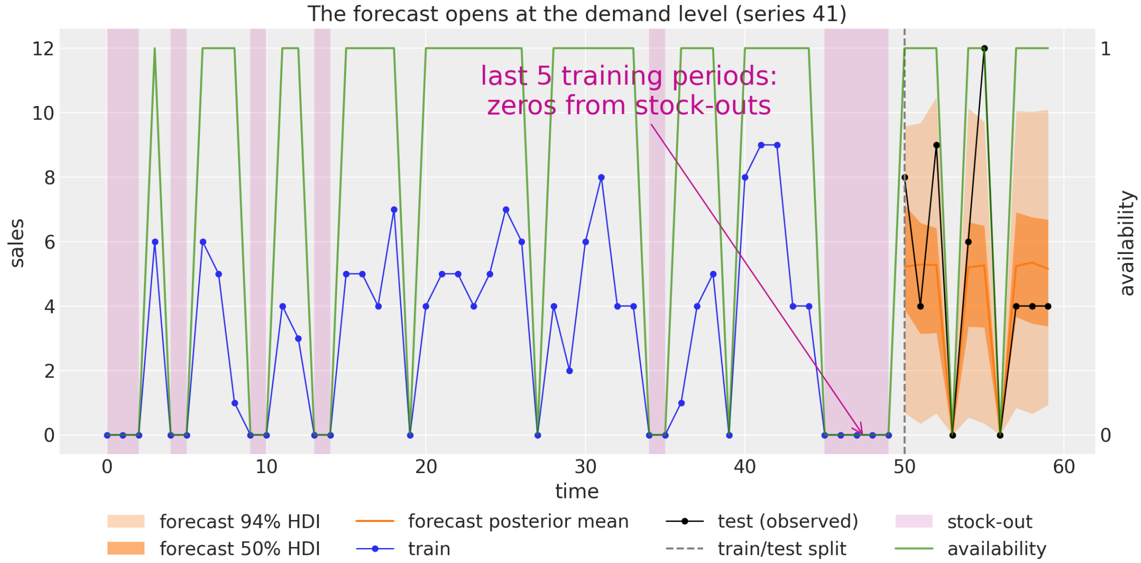

Figure 4: A series whose training window ends in a five-period stockout run. The full-availability demand forecast (bands: 50% and 94% HDI) opens at the inferred demand level instead of at the trailing zeros; the realized-availability forecast (left of the split) pinches to zero inside stockouts.

4.2 Filtered lags

The mean model is an autoregression with weekly seasonality,

$$\hat{y}_t = \mu + \phi_1 \tilde{y}_{t-1} + \phi_2 \tilde{y}_{t-2} + \mathbf{f}_t^\top \boldsymbol{\beta}, \tag{4}$$

with $\mathbf{f}_t$ a weekly Fourier basis. An autoregression on stockout-affected data has a second entry point for corruption beyond the likelihood: the lags themselves. Feeding raw zeros from stockout days as lagged sales attenuates ϕ_1, ϕ_2 toward zero and inflates σ, because the first on-shelf day after a gap looks like an impossible rebound. The filtered lag

$$\tilde{y}_t \;=\; a_t \left[(1 - c_t^{\text{ind}})\, y_t + c_t^{\text{ind}} \, \max(y_t, \hat{y}_t) \right] + (1 - a_t)\, \hat{y}_t \tag{5}$$

passes clean on-shelf observations through, floors capped observations at the model's own prediction, and substitutes the prediction on off-shelf days, the plug-in analogue of how a state-space filter handles missing observations. The $\max(y_t, \hat{y}_t)$ term is a plug-in approximation to the exact censored conditional mean $\hat{y}_t + \sigma\phi(z_t)/(1 - \Phi(z_t))$; the approximation is deliberate, cheap, and one of the acknowledged simplifications below. Over the forecast horizon the recursion feeds sampled trajectories back into the carry and never applies a cap, which is how this case study obtains its demand forecast (Table 2).

4.3 Setup

The generator produces 180 periods of latent demand from an AR(2) with weekly seasonality ($\phi_1 = 0.6$, $\phi_2 = 0.3$), subtracts a small sales friction $\delta = 0.25$, caps at a constant capacity $c = 2.2$, and gates through Bernoulli(0.8) availability: 49 stockout days (27%) and 36 capacity-censored days (20%), with the last 30 days held out. The friction term is absent from the fitted model, so the model is again misspecified relative to the generator (§2.3). Inference is the No-U-Turn Sampler

Simulated panel, 1,000 × 60*; SVI; different data and protocol from Table 3; not comparable to it or to later tables.*		
	Availability-gated TSB	Plain TSB (all-ones refit)
Mean estimated / true demand probability	0.99	0.60
Full-availability panel total (true 2,471.8)	2,481.3	1,479.8
Panel test CRPS (realized availability)	0.547	
Coverage 50% / 94% (realized availability)	0.70 / 0.98	

Table 4: Availability-gated TSB versus the plain-TSB ablation on the simulated panel. The 0.60 recovery ratio equals the generator's availability rate by construction (see text); the comparison is a consistency check on the gating mechanism. CRPS and coverage for the plain refit are omitted because under realized availability the two sales forecasts are close and the difference is not the point of the comparison.

with 4 chains of 1,000 warmup and 1,000 draws each; all $\hat{R}$ values are 1.00. Figure 5 shows the anatomy of the corruption.

The comparison arm refits the same model with the censoring indicator zeroed out, under which (3) reduces to a plain Normal likelihood that treats capped days as exact observations of 2.2. Two properties of this ablation must be stated to keep the comparison honest. First, zeroing the indicator changes *two* coupled things: the likelihood loses its survival term, and the lag filter (5) loses its cap-flooring, so the ablation measures the joint effect of both. Second, both arms keep the availability masking and off-shelf lag substitution, so the baseline is a *plain Normal likelihood* model that already handles stockouts, not a naive fit to raw sales; the comparison isolates the treatment of capacity censoring specifically.

4.4 Results: three questions

(a) Aggregate accuracy against latent demand. On the full 30-day holdout scored against the latent demand path, the censored model does not win: the plain-Normal model has lower RMSE (0.572 versus 0.635) and lower CRPS (0.333 versus 0.370), while MAE goes the other way (0.495 versus 0.488 for the censored model). The two models genuinely agree on most days: roughly two thirds of the test days have demand below the cap, where the likelihoods coincide, and the plain model's tighter predictive scores well there. Aggregate point accuracy on a mostly-uncensored window is the wrong place to look for the mechanism.

(b) Stress days and calibration. The models differ exactly where the ordering decision lives. On the 10 of 30 test days where latent demand exceeds the cap, the censored model's MAE is 0.318 against 0.687 for the plain model, less than half the error, with a stated mechanism: the plain model's conditional mean tops out near the cap it mistook for data, while the survival term lets the censored model place mass above it. Figure 6 shows the same behavior in-sample, with the censored model's HDI bands riding above the cap on capacity days. Interval calibration moves in the same direction, 0.600 versus 0.467 empirical coverage at nominal 50% and 1.000 versus 0.967 at 94%, but on 30 days these gaps are at most directional (§2.3): the 50% gap is within about 1.5 binomial standard errors and the 94% gap is a single day. The peak-day contrast, not the coverage table, is this case study's headline.

(c) The trap, demonstrated. Scored against *observed sales* instead of latent demand, the ranking inverts: the plain model wins every metric, for example CRPS 0.774 versus 0.991. This is the outcome predicted in §2.3: the plain model is rewarded precisely for reproducing the caps and the

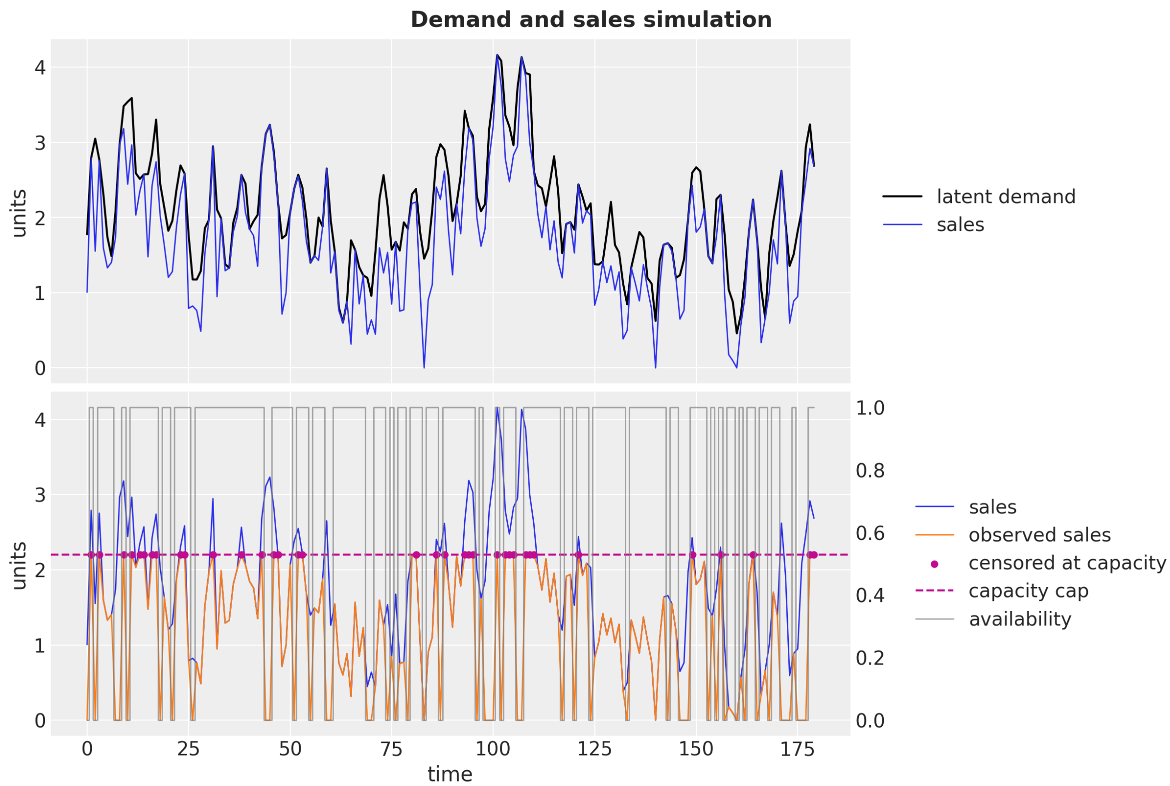

Figure 5: The simulated censoring anatomy. Top: latent demand versus sales after friction. Bottom: sales versus observed sales, with stockout gaps, the capacity cap (dashed), and capacity-censored days marked. Planning needs the black curve; the data records the orange one.

post-stockout dynamics that the censored model refuses to reproduce. A model selection exercise run on recorded sales would have chosen the model that systematically under-forecasts demand on the days that cause stockouts. Aggregate scores against sales evaluate the censoring process, not the demand model.

Limitations. Exact parameter recovery is not achieved and is not achievable here: the recursion runs on filtered sales rather than demand, and the generator's friction and zero-clipping are unmodeled (ϕ_1 posterior mean 0.453 against a true 0.6). The method requires trusting the recorded censoring indicator and observing availability; in this simulation censoring is detected by exact equality with the cap, which is an artifact of the generator, and production data would need a threshold rule. The Normal likelihood places predictive mass on negative sales in-sample. The lag filter is a plug-in approximation, the cap is constant where real shelf capacity varies, and all results come from a single fixed split with no rolling-origin backtest.

5 Case study 3: fractional availability at 50,000 series

The first two case studies used simulated data so that recovery of known latent quantities could be checked. The third works on real data at production scale, where the stockout signal is neither a clean flag nor a known cap but a *fractional, noisily labeled* availability, and where the modeling choices are dominated by what survives contact with label noise and a large panel.

Single simulated series, 30-day holdout, NUTS; not comparable to the tables of §3 or §5.						
Scored against	Model	MAE	RMSE	CRPS	Cov. 50%	Cov. 94%
Latent demand	censored	0.488	0.635	0.370	0.600	1.000
Latent demand	plain Normal	0.495	0.572	0.333	0.467	0.967
Latent demand, peak days (n=10)	censored	0.318	0.341	0.274	0.8	1.0
Latent demand, peak days (n=10)	plain Normal	0.687	0.744	0.458	0.3	0.9
Observed sales	censored	1.334	1.616	0.991	0.300	0.600
Observed sales	plain Normal	0.996	1.204	0.774	0.267	0.467

Table 5: Censored versus plain-Normal likelihood on the capacity case study. The ablation flips one input but changes two coupled things (survival term and lag cap-flooring); both arms retain stockout masking. Coverage gaps on 30 days (binomial standard error ≈ 0.09 at nominal 50%) are directional evidence only. Peak days are the 10 of 30 test days with latent demand above the cap.

5.1 Data and label noise

FreshRetailNet-50K (Wang et al., 2025) contains 90 days of daily sales for 50,000 store-product series (4,500,000 rows, 898 stores, 865 products, 18 cities, 2024-03-28 to 2024-06-25), with hourly stockout flags per day. We aggregate the hourly flags into a sales-weighted availability $a_{t,s} = \sum_h w_h(1 - \text{stockout}_{t,s,h})$, where w_h is the share of total sales occurring in hour h, so that an item off shelf only overnight loses little availability. The last 14 days are held out; the first 76 train.

The stockout labels are visibly noisy, and the noise shapes the model. On days flagged out of stock for all 24 hours, 14.8% nonetheless record positive sales; 2.5% of all sales volume occurs in hours flagged out of stock; and the empirical sales ratio at zero availability is 0.037 rather than zero. A mechanism that forces sales to zero at zero availability would be contradicted by the data it is fit to. This motivates the learned floor below.

5.2 Model

Each series carries a local-level state-space model with a damped trend, weekly seasonality, and store-hierarchical covariate effects, all multiplied by an availability factor:

$$y_{t,s} \sim \text{Normal}(f_{t,s}\,\mu_{t,s},\; f_{t,s}\,(\sigma_s + \lambda_s\,\text{softplus}(\ell_{t,s})) + \sigma_0)\,, \tag{6}$$

$$\mu_{t,s} = \ell_{t,s} + \gamma_{d(t),s} + \textstyle\sum_{c=1}^{4} \beta_{c,s}\,x_{c,t,s}, \tag{7}$$

$$\ell_{t,s} = \ell_{0,s} + \textstyle\sum_{u\le t}(\varepsilon_{u,s} + \delta_{u,s}), \qquad \varepsilon_{u,s} \sim \text{Normal}(0, \tau_s), \tag{8}$$

$$\delta_{u,s} = \phi_s^{\text{trend}}\,\delta_{u-1,s} + \eta_{u,s}, \qquad \eta_{u,s} \sim \text{Normal}(0, \tau_s^{\text{trend}}), \tag{9}$$

where $\gamma_{d(t),s}$ is a zero-sum weekly seasonal effect, the covariates $x_{c,t,s}$ are discount magnitude, promotional activity, holiday, and a launch indicator, and the coefficients $\beta_{c,s}$ shrink toward store-level means. The availability factor is

$$f_{t,s} \;=\; \phi_s + (1 - \phi_s)\,\frac{1 - e^{-b_s a_{t,s}}}{1 - e^{-b_s}}, \tag{10}$$

with $\phi_s \sim \text{Beta}(2, 18)$ and $b_s \sim \text{LogNormal}(1, 0.5)$. Three properties make (10) more than a fudge factor. The floor ϕ_s gives zero-availability days a small positive expected sale, absorbing the label noise quantified above instead of letting contradiction days dominate the likelihood; its prior is concentrated near the empirical floor of 0.037, so posterior agreement with that value is partly

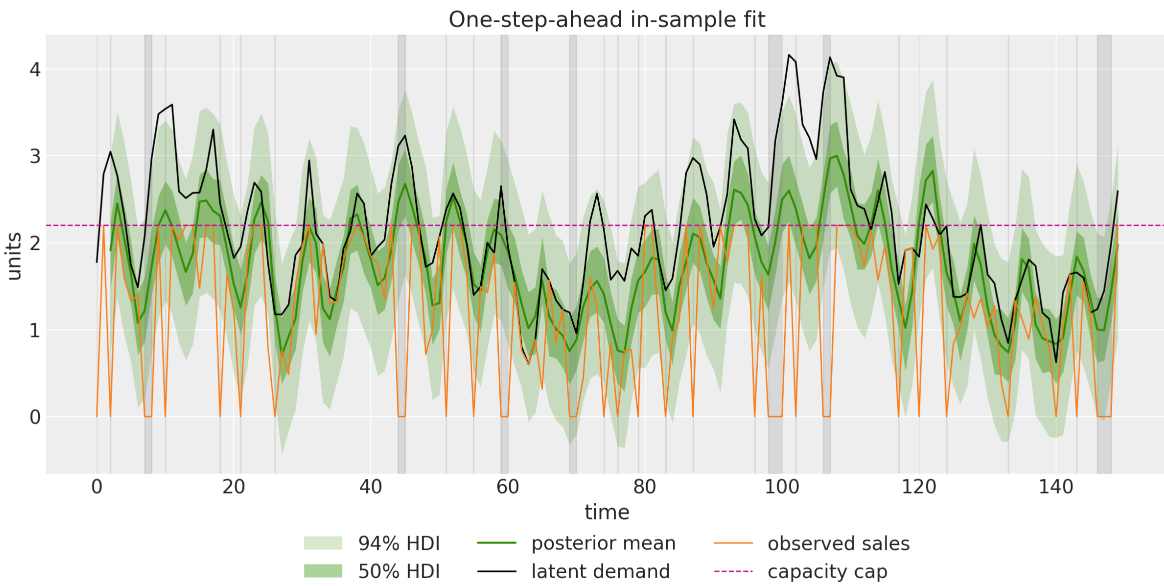

Figure 6: In-sample one-step-ahead fit of the censored model: the 50% and 94% HDI bands ride above the cap on capacity-censored days and carry the demand estimate through shaded stockout gaps, tracking the latent demand overlay.

prior-driven and should be read as consistency, not as independent validation. The saturating shape has a random-encounter interpretation: if customers make purchase attempts at Poisson rate b_s and succeed only when the item is in stock a share a of the day, the probability of at least one successful encounter is $1 - e^{-b_s a}$. The normalization by $1 - e^{-b_s}$ anchors $f_{t,s} = 1$ at full availability, which removes the scale degeneracy between the level and the factor; at $a = 1$ the observation mean is the full, uncorrupted regression mean $\mu_{t,s}$ (level plus seasonality plus covariate effects). The factor also multiplies the observation noise scale in (6), so low-availability days are both smaller in mean and less informative, while a small constant $\sigma_0 = 0.02$ outside the factor keeps the likelihood proper at $a = 0$. Sales are scaled per series by the training-window mean before fitting.

5.3 Computation

The model is fit with stochastic variational inference under an `AutoNormal` guide for 60,000 steps, using a one-cycle learning-rate schedule combined with reduce-on-plateau. The notebook's stored timing for this fit on the full 50,000-series panel is 5 minutes 28 seconds of wall-clock time with the CUDA backend. The committed companion notebook in the numpyro_forecast repository, which runs the identical model on a 1,000-series subset on CPU, reports that a hosted version of the full-panel workflow (provisioned via Modal[1]) executes end to end, from data loading through counterfactual forecasting, in approximately 10 minutes on GPU; we quote that as the notebook's statement rather than an independent measurement. The bounded claim these numbers support is that the same hierarchical specification used on the 1,000-series subset runs at the 50,000-series scale in minutes on one machine; posterior-predictive export at this scale requires draw batching and host offload, which the package provides.

[1] https://modal.com

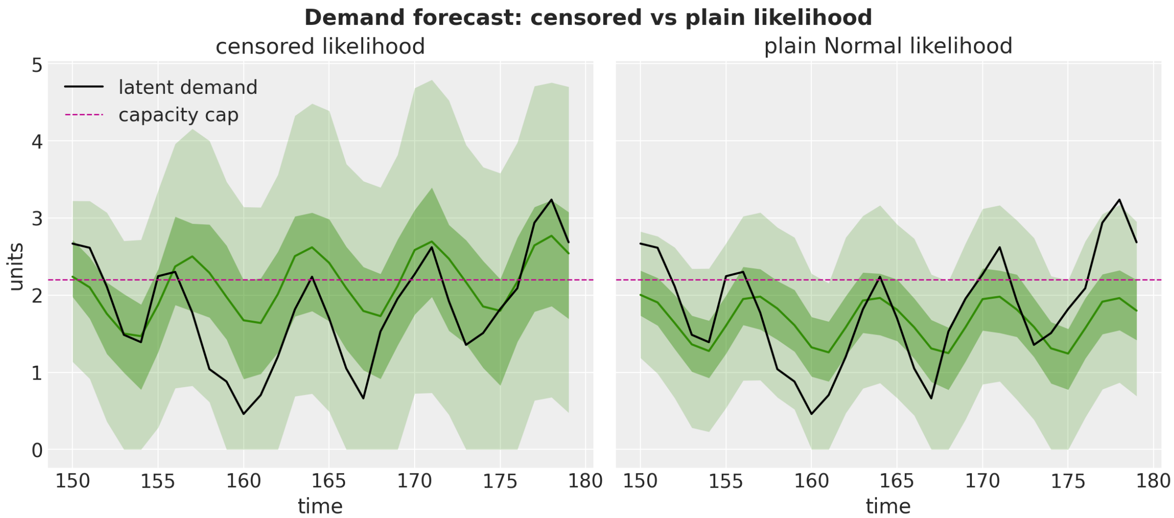

Figure 7: Thirty-day demand forecasts of the censored model (left) and the plain-Normal ablation (right) against latent demand, with the capacity cap dashed. The models agree below the cap; the plain model's central tendency saturates near the cap on the days demand exceeds it.

5.4 Results

On the 14-day holdout, scored on the original sales scale, the model's CRPS is 0.268 against 0.380 for a seasonal-naive ensemble (weekday-aligned training windows stacked into an empirical predictive); central-interval coverage is 0.938 at nominal 94% and 0.556 at nominal 50%. Two attribution caveats belong in the same breath as these numbers. The seasonal-naive comparison scores the model *as a whole*, trend, seasonality, promotions, hierarchy, and availability factor together; because no with-and-without-factor ablation was run at this scale (or any scale), none of the CRPS margin can be attributed to the availability factor specifically. And coverage degrades on zero-sales days (0.36 at nominal 50%, against 0.56 on positive days; zeros are 3.7% of the test panel), direct evidence of the Normal likelihood's misspecification on clipped, non-negative sales.

The demand forecast pins availability to one over the horizon and re-forecasts with the same posterior draws and the same random numbers, so the difference between the two forecasts is the model's implied stockout cost, free of Monte Carlo noise. Across the panel the fitted model implies 8.7% more expected demand than expected sales over the test window, with 82.2% of series gaining more than 1%; this is a model output, not a validated quantity, since demand at full availability is never observed, and the availability labels it conditions on are themselves noisy. Figure 9 shows one illustrative series: on its worst test day (availability 0.42), expected sales are 13.9 units against an expected demand of 25.0.

Limitations. All results use a single train-test split. Availability over the horizon is taken from the recorded data, so the evaluation is retrospective; a production system would need availability forecasts or scenarios. The full-availability demand forecast is unfalsifiable against observations by construction. The Normal likelihood on non-negative, zero-clipped sales is misspecified, with the zero-sales-day coverage gap above as its measurable symptom, and predictive intervals become directionally miscalibrated (systematically missing high) in the second forecast week. The sales-weighted hour weights w_h were computed over the full window including test days, a mild hygiene compromise inherited from the source notebook. The hosted end-to-end runtime is reported

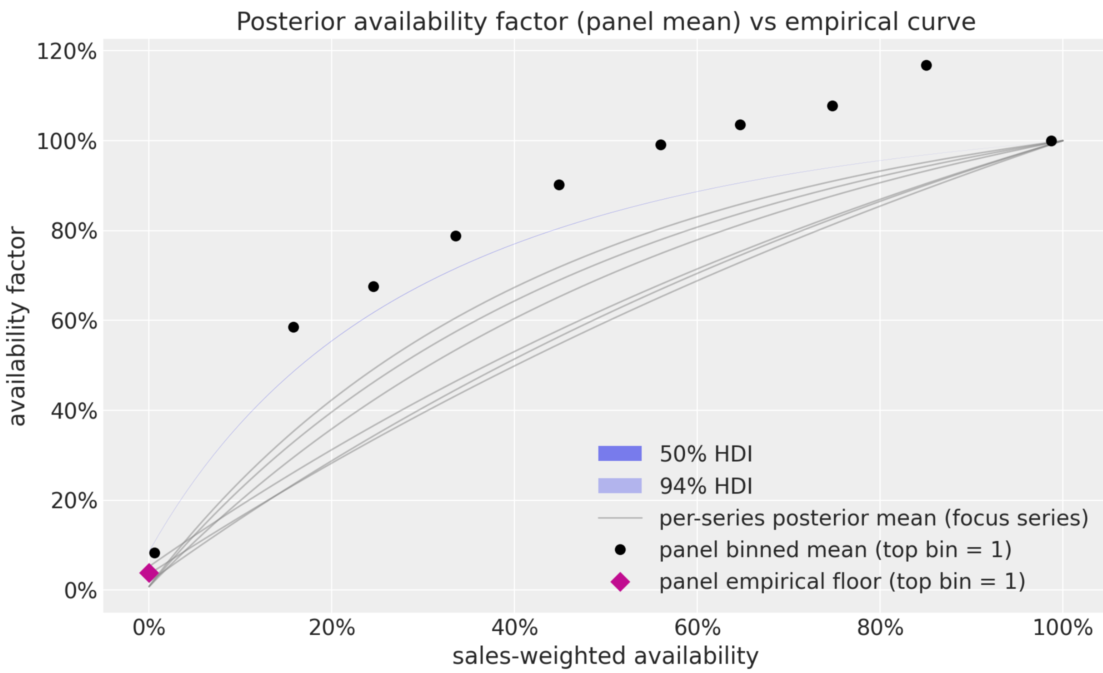

Figure 8: Posterior panel-mean availability factor against availability, with 50% and 94% HDI bands, overlaid on the rescaled empirical sales ratios by availability decile. The posterior curve sits below the raw empirical points over most of the range, consistent with the model attributing part of the high sales on low-availability days to those days being high-demand days (stockouts are not random). The bands quantify uncertainty of the panel *average* and are accordingly thin; per-series factors vary far more.

secondhand, as described in §5.3.

6 Discussion

Shared structural assumptions. Three assumptions run through all three case studies and deserve to be stated once, prominently. First, *availability is treated as exogenous*: the models condition on the stockout process but do not model why stockouts happen. In reality stockouts concentrate on high-demand days and best-selling items, so the censoring is informative; Braden and Freimer (1991) show how censoring changes what observations teach, and the choice-based literature (§2.4) treats the demand-substitution side our single-item models ignore. The fresh-retail factor curve sitting below the raw empirical sales ratios (Figure 8) is a visible trace of this endogeneity inside a model that does not represent it explicitly. Second, *Gaussian likelihoods are used throughout as a pragmatic compromise*, on counts, on a binary indicator, and on non-negative clipped sales; each case study reported a measurable symptom (negative predictive draws, probability draws outside $[0, 1]$, degraded coverage on zero-sales days). Count and truncated likelihoods are the natural refinements, at some computational cost at scale. Third, *availability itself must come from somewhere* at forecast time: the case studies either assume it (scenario inputs) or take it from recorded data (retrospective evaluation); forecasting availability is its own problem and nothing here solves it.

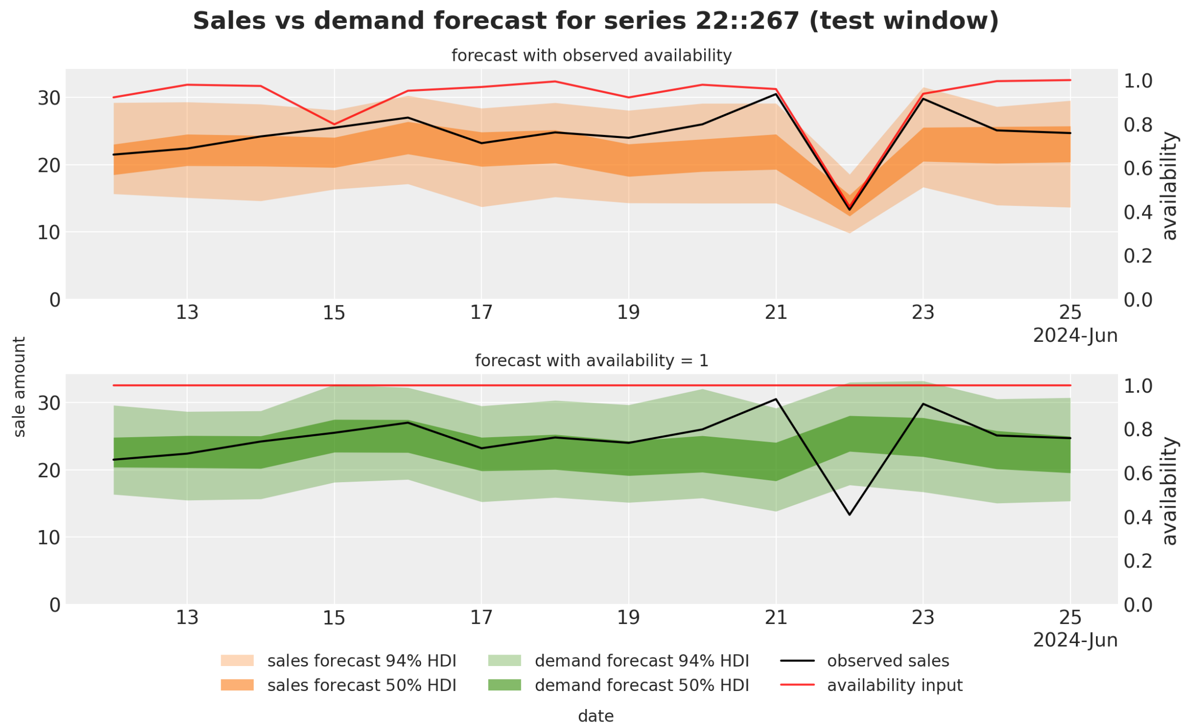

Figure 9: Sales forecast under observed availability (top) versus demand forecast under availability pinned to one (bottom) for one illustrative series, with 50% and 94% HDI bands and the availability input in red. On 2024-06-22 (availability 0.42) the expected sale is 13.9 units and the expected demand 25.0. The series was chosen to display the mechanism clearly, not as a typical case.

What a controlled comparison would require. This paper characterizes mechanisms and their qualitative failure modes; it does not rank them. A ranking would require a dataset recording *both* a binary or fractional availability signal and a per-observation censoring point on the same series, so that gating, censored likelihoods, and multiplicative factors could be cross-applied; a with-and-without-factor ablation at the 50,000-series scale; rolling-origin evaluation rather than the single splits used here; and stress-day scoring defined independently of any candidate model. Until then, claims of the form "mechanism X beats mechanism Y" are not supported by the evidence in this paper, and we have tried not to make them.

When each mechanism shades into the next. The taxonomy of Table 2 has soft edges. A binary flag is the degenerate case of a fractional availability, and the gate $g_t = a_t$ is what the factor (10) becomes when $a_t \in \{0, 1\}$, the floor is zero, and the model is a recursion rather than a regression. A recorded inventory level is simultaneously a censoring point (when sales hit it) and an availability signal (when it is zero), so the mechanisms compose: one can gate a recursion on the zeros and censor the likelihood at the recorded stock. The practical reading of the taxonomy is not "pick exactly one row" but "let what is recorded decide which components the likelihood can honestly contain."

FreshRetailNet-50K, 14-day holdout, original sales scale; not comparable to the tables of §3 or §4.			
	CRPS	Coverage 94%	Coverage 50%
Hierarchical model with availability factor	0.268	0.938	0.556
Seasonal-naive ensemble	0.380	–	–
Fit wall-clock: 5 min 28 s (60,000 SVI steps, 50,000 series, CUDA backend).			

Table 6: Panel evaluation for the fresh-retail case study. The seasonal-naive baseline is an empirical ensemble without calibrated intervals, so its coverages are not defined. No availability-factor ablation is available: the CRPS margin reflects the whole model, not the factor.

7 Reproducibility

All three case studies are executable examples in numpyro_forecast[2], an open-source JAX (Bradbury et al., 2018) and NumPyro (Phan et al., 2019; Bingham et al., 2019) library for probabilistic time-series forecasting that ports the design of Pyro's forecasting module. Section 3 corresponds to the `availability_tsb` example (with `croston` and `tsb` as companions), §4 to `censored_demand`, and §5 to `fresh_retail_stockout`. Every number quoted in this paper is a stored output of those notebooks; the figures are extracted directly from the committed notebook outputs by a script in the paper's source tree, without re-executing the notebooks. All examples use fixed random seeds. One disclosure: the 50,000-series GPU variant of the fresh-retail notebook, from which the §5 results are taken, is not distributed in the repository at the time of writing; the committed notebook covers the identical model on a 1,000-series subset on CPU and links to the hosted full-panel run. The exponential smoothing state-space background for the models of §3 is in Hyndman et al. (2008); the state-space machinery of §5 follows Durbin and Koopman (2012); general forecasting practice is covered by Hyndman and Athanasopoulos (2021), and CRPS is the proper scoring rule of Gneiting and Raftery (2007), reviewed in Gneiting and Katzfuss (2014).

8 Conclusion

Recorded sales are a censored measurement of demand, and models fit to raw sales inherit the censoring: they learn that stockout days sell nothing and capped days sell the cap, and then plan accordingly. The three case studies in this paper worked one repair each, selected by what the data records about the stockout. A binary on-shelf flag lets a smoothing recursion freeze its occurrence updates off shelf, so that a stockout run neither erodes the demand estimate nor drags the reopening forecast to zero. A per-observation cap lets a censored likelihood score capped days as lower bounds, which cut peak-day errors by half in our simulation precisely where the plain likelihood saturated at the cap. A fractional, noisily labeled availability supports a multiplicative saturating factor whose learned floor absorbs the label noise, fit here to 50,000 series in minutes on a GPU. All three share one structure: the corruption is an explicit, switchable component of the model, and the demand forecast is the model run with the component switched off. And all three were evaluated under the same discipline: aggregate accuracy against recorded sales rewards reproducing the corruption, so demand models must instead be judged on calibration, stress-day behavior, and the plausibility of their counterfactuals. The mechanisms are simple, composable, and implemented in an open-source package; what remains genuinely open, endogenous availability, substitution, count likelihoods at

[2] `https://github.com/juanitorduz/numpyro_forecast`, documentation at `https://juanitorduz.github.io/numpyro_forecast/`.

scale, and the controlled cross-mechanism comparison, is catalogued in §6.

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